7. The angular momentum of a moving body remains constant if

(1) Net external force is zero

(2) Net pressure is applied

(3) Net external torque is applied

(4) Net external torque is zero

8. Keeping moment of inertia constant, the angular momentum of a body is increased by 20%. The percentage increase in its rotational kinetic energy is

(1) 40%

(2) 12%

(3) 44%

(4) 20%

9. A solid body rotates about a stationary axis so that the angular positon θ varies with time as $\theta = 6t - 2t^3$ radian. The angular acceleration at the moment when the body stop is (Here t is in second)

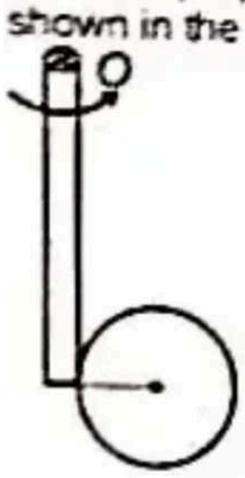
(1) 6 rad/s^2

(2) 12 rad/s^2

(3) -12 rad/s^2

(4) Zero

10. A uniform rod of length $2\sqrt{2}\ell$ and mass m are welded with a disc of mass m and radius ℓ. The radius of gyration of the system, about an axis passing through one end of rod and perpendicular to the plane of rod and disc as shown in the figure, will be



(1) $\ell\sqrt{\frac{22}{3}}$

(2) $\ell\sqrt{\frac{22}{12}}$

(3) $\ell\sqrt{\frac{14}{3}}$

(4) $\ell\sqrt{\frac{22}{3}}$

11. The radius of gyration of a hollow sphere of radius r, about a certain axis is r. The distance of this axis from the centre of sphere is

(1) $\sqrt{\frac{61}{19}}\,r$

(2) $\sqrt{12}\,r$

(3) $\sqrt{5}\,r$

(4) $\frac{r}{\sqrt{3}}$

12. The radius of gyration of a uniform rod of mass and length l about an axis perpendicular to its length and at a distance $\frac{l}{4}$ from its one end will be

(1) $\frac{l}{2}\sqrt{\frac{7}{3}}$

(2) $\ell\sqrt{\frac{7}{3}}$

(3) $\frac{\ell}{4}\sqrt{\frac{7}{3}}$

(4) $2\ell\sqrt{\frac{7}{3}}$

13. A wheel is at rest. Its angular velocity increases uniformly and becomes 50 rad/s after 10 s. The total angular displacement in 10 s is

(1) 100 rad

(2) 150 rad

(3) 250 rad

(4) 300 rad

14. The dimensional formula of "Moment of force" is

(1) $[M^1L^1T^{-2}]$

(2) $[M^1L^2T^{-1}]$

(3) $[M^1L^2T^{-2}]$

(4) $[M^1L^1T^{-1}]$

15. Two point masses of mass **1** kg and 2 kg are located at coordinates (1 m, 2 m) and (2 m, 1 m) respectively. The moment of inertia of this system about x-axis is equal to

(1) 12 kg m^2

(2) 8 kg m^2

(3) 4 kg m^2

(4) 6 kg m^2

16. Select the correct option regarding moment of inertia of a point mass.

(1) It changes if the mass is shifted parallel to the axis of the rotation

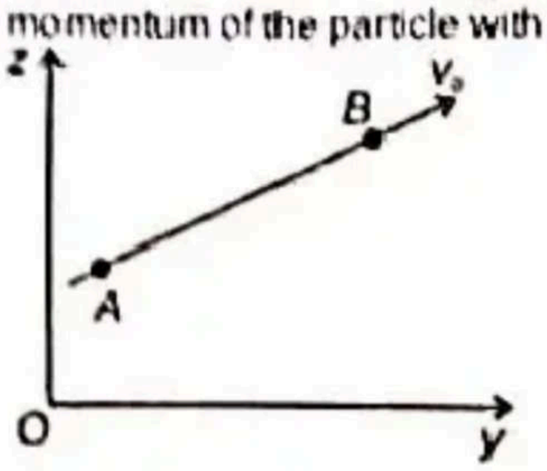
(2) It depends on the distance of the mass from the axis of rotation

(3) It does not change, if the mass is rotated with variable angular velocity about axis of rotation

(4) Both (2) & (3)

EinsteinXSeg

A particle of mass m is moving in the y-z plane with a velocity v_0 along the straight line AB. The angular momentum of the particle with respect to origin O



(1) Will increase

(2) Will decrease

(3) Remains constant

(4) Will be zero

18. A man stands at one end of a boat which is stationary in water. Neglect water resistance. The man now moves to the other end of the boat and again becomes stationary. The centre of mass of the 'man plus boat' system will remain stationary with respect to water

(1) Only when the man is stationary initially and finally

(2) Only if the man moves without acceleration on the boat

(3) Only if the man and the boat have equal masses

(4) In all cases

19. A particle is moving along line $y = 4x$ with a speed 10 m/s. If the mass of the particle is 2 kg then its angular momentum about the point (1, 4) is

(1) 10 kg m^2/s

(2) 20 kg m^2/s

(3) Zero

(4) 5 kg m^2/s

20. A force with a given magnitude is to be applied to a wheel. The torque can be maximised by

(1) Applying the force near the axle, radially outward from the axle

(2) Applying the force near the rim, radially outward from the axle

(3) Applying the force near the axle, parallel to a tangent to the wheel

(4) Applying the force at the rim, tangent to the rim

21. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $7A_0$ in 3 s. The magnitude of average torque is

(1) $6A_0$

(2) $3A_0$

(3) $\frac{3A_0}{4}$

(4) $2A_0$

22. A sphere is rotating about its diameter with uniform angular velocity, then

(1) Acceleration of particles on surface of sphere is zero

(2) Acceleration of particles at a distance half of radius from axis is zero

(3) All particles have same linear speed

(4) Particles may have different linear speeds

23. A body is in pure rotation. The linear speed v of a particle, the distance r of the particle from the axis of rotation and the angular velocity are related as $\omega = \frac{v}{r}$. Thus,

(1) $\omega \propto \frac{1}{r}$

(2) $\omega \propto r$

(3) $\omega \propto r^2$

(4) ω is independent of r

24. A shaft rotating at 3000 rpm is transmitting a power of 3.14 kW. The magnitude of the driving torque is

(1) 6 N m

(2) 10 N m

(3) 15 N m

(4) 22 N m

25. A : Internal forces can change the kinetic energy but not the linear momentum of the system.
R : Net internal force on the system is non-zero.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

26. A thin uniform rod of mass 1 kg and length 0.3 m is smoothly hinged at one end. The rod is released from horizontal position and allowed to rotate in vertical plane. The angular velocity of the rod when it becomes vertical is ($g = 10$ m/s^2)

(1) $5\sqrt{3}$ rad/s

(2) 10 rad/s

(3) $10\sqrt{3}$ rad/s

(4) $\frac{10}{\sqrt{3}}$ rad/s

27. On applying couple on a rigid rod initially at rest, it will

(1) Produce pure linear motion

(2) Produce pure rotational motion

(3) Produce linear and rotational motion

(4) Be in equilibrium

Einstein

28. A system consists of two identical particles. One particle is at rest and other particle has velocity v. The centre of mass of the system has a velocity of

(1) $\frac{v}{2}$

(2) v

(3) $2v$

(4) $\frac{v}{4}$

29. Select the correct option about moment of inertia.

(1) It is same for all the axes

(2) Depends on distribution of mass from axis of rotation

(3) Theorem of perpendicular axes is applicable for all type of bodies

(4) Both (2) and (3)

30. A wheel initially has an angular velocity of 18 rad/s. It has a constant angular acceleration of 2 rad/s^2 and is slowing at first. What time elapses before its angular velocity is 22 rad/s in the direction opposite to its initial angular velocity?

(1) 2 s

(2) 20 s

(3) 1 s

(4) 10 s

31. Two persons of masses 60 kg and 80 kg respectively, are at the opposite ends of a boat. The length of the boat is 4.0 m and weighs 100 kg. Both persons move towards each other and sit in the middle of the boat. The centre of mass of the system shifts by (neglect water friction)

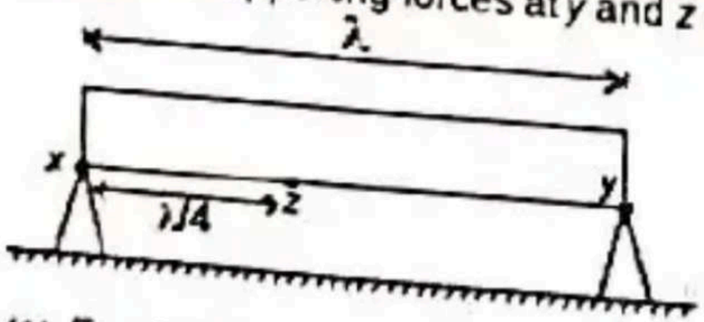
(1) 0.25 m

(2) 0.5 m

(3) 0.75 m

(4) zero

32. A uniform plank is supported by two equal 120 N forces at x and y as shown in figure. If the support at x is moved to z, then the supporting forces at y and z are



(1) $F_y = 120$ N; $F_z = 120$ N

(2) $F_y = 40$ N; $F_z = 200$ N

(3) $F_y = 60$ N; $F_z = 180$ N

(4) $F_y = 80$ N; $F_z = 160$ N

33. A meter stick with uniform linear mass distribution is balanced at its centre. Now a small block of mass [illegible] is put at 16 cm mark of scale. The stick is found balanced at 40 cm mark. The mass of meter stick is

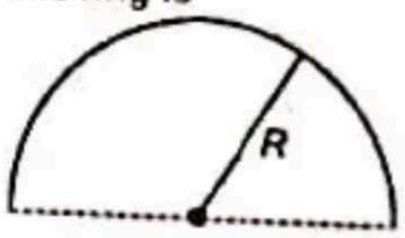
(1) 4.3 kg

(2) 5.6 kg

(3) 7.0 kg

(4) 9.6 kg

34. A semicircular ring of mass M and radius R is as shown in figure. Radius of gyration of ring about an axis passing through centre of curvature and perpendicular to plane of this ring is



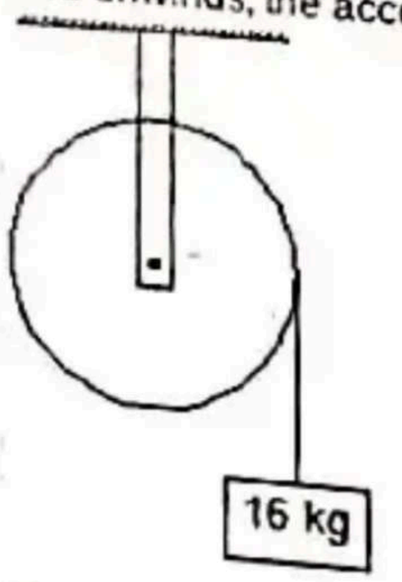
(1) $\frac{1}{\sqrt{2}}R$

(2) $\frac{R}{2}$

(3) R

(4) $\sqrt{2}R$

35. A 16 kg block is attached to a cord that is wrapped around the rim of a flywheel of diameter 0.4 m and hangs vertically, as shown in figure. The rotational inertia of the flywheel is 0.5 kg m^2. When the block is released and the cord unwinds, the acceleration of the block is



(1) g

(2) 0.84 g

(3) 0.56 g

(4) 0.14 g

36. The moment of inertia of a uniform cylinder about an axis is $\frac{5MR^2}{2}$. The value of radius of gyration of the cylinder about this axis will be (M and R are mass and radius of cylinder)

(1) $\sqrt{\frac{5}{2}}R$

(2) $\frac{5}{\sqrt{2}}R$

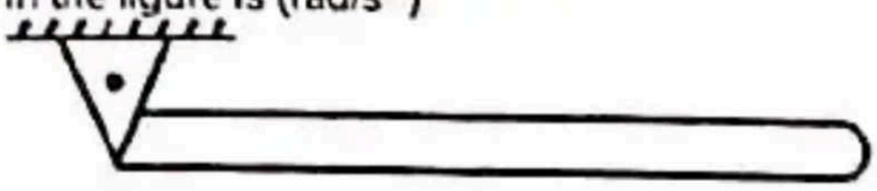
(3) $\frac{5}{\sqrt{2}}R$

(4) $\frac{R}{\sqrt{2}}$

Einstein 369

distribution
mass 4 kg
d to be

A uniform rod of mass 10 kg and length 3 m is pivoted at its end. Angular acceleration of rod about pivot, just after it is released from rest in the horizontal position as shown in the figure is (rad/s^2)



(1) 2

(2) $\sqrt{2}$

(3) $\sqrt{5}$

(4) 5

38. A body experiences a torque which acted for only $\frac{1}{2}$ s. If during this time, torque (τ) varied as, $\tau = (3t^2 + 2t)$ Nm, where t is time in second. Then angular impulse acted on the body for this small interval is

(1) $\frac{1}{8}$ kg m^2 s^{-1}

(2) $\frac{3}{4}$ kg m^2 s^{-1}

(3) $\frac{3}{8}$ kg m^2 s^{-1}

(4) $\frac{5}{3}$ kg m^2 s^{-1}

39. **Assertion (A)** : When net external torque on a body is zero, then its linear momentum must remains constant.
Reason (R) : When net force acting on a body is zero, then its angular momentum must remain constant.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) Both A and R are false

40. A system consists of two particles of masses m and M ($M > m$). The centre of mass of the system is

(1) At the middle

(2) Nearer to M

(3) Nearer to m

(4) At the position of large mass

41. A disc is rotating at a constant angular velocity of 300 rpm. The angular displacement of disc in one second is

(1) $\frac{3}{50\pi}$ rad

(2) $\frac{3\pi}{50}$ rad

(3) $\frac{25\pi}{3}$ rad

(4) 10π rad

42. A 500 kg boat is 9 m long and is floating without motion on still water. A man of mass 100 kg is at one end and move to other end of boat and stops. The displacement of boat is

(1) 1.5 m in the direction of displacement of man

(2) 0.75 m in the direction of displacement of man

(3) 1.5 m in the direction opposite to displacement of man

(4) 0.75 m in the direction opposite to displacement of man

43. From a complete ring of mass M and radius R, an arc of length $\frac{\pi R}{3}$ is removed. The moment of inertia of the incomplete ring about an axis passing through the centre of ring and normal to the plane of ring, is

(1) $\frac{11}{12}MR^2$

(2) $\frac{MR^2}{18}$

(3) MR^2

(4) $\frac{5}{6}MR^2$

44. Moment of inertia of a semicircular disc of mass M and radius R about an axis passing through centre of curvature and perpendicular to the plane of disc is

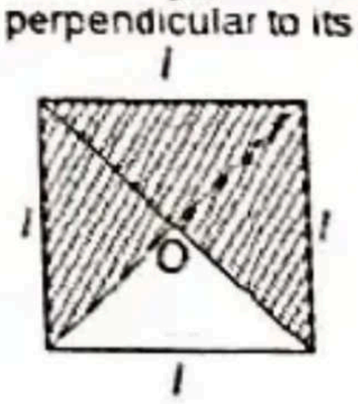
(1) $\frac{2MR^2}{3}$

(2) MR^2

(3) $\frac{2MR^2}{5}$

(4) $\frac{MR^2}{2}$

45. An isosceles triangular piece is cut from a square plate of side l. The piece is one-fourth of the square and mass of the remaining plate is m. The moment of inertia of the remaining plate about an axis passing through O and perpendicular to its plane is



(1) $\frac{ml^2}{6}$

(2) $\frac{ml^2}{12}$

(3) $\frac{ml^2}{24}$

(4) $\frac{ml^2}{3}$

EinsteinX869

Chemistry

46. pH of a solution containing 1 mol of CH_3COOH and 0.2 mol of CH_3COONa is [given pK_a of CH_3COOH is 4.76]

(1) 5.76

(2) 4.76

(3) 4.06

(4) 5.46

47. If for the reaction $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$, K_C is 2×10^{10}. then the K_P for the reaction at the same temperature T K is

(1) 2×10^{10}

(2) 2×10^{10} RT

(3) $\frac{RT}{2 \times 10^{10}}$

(4) $\frac{2 \times 10^{10}}{RT}$

48. If the equilibrium constant for,
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is K, then the equilibrium constant for $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$ is

(1) $\sqrt{K}$

(2) K^2

(3) $\frac{1}{\sqrt{K}}$

(4) $\frac{1}{2}K$

49. If the rate constant of forward reaction is triple of rate constant of backward reaction under identical condition at equilibrium, then K_{eq} for the reaction is

(1) 0.33

(2) 3.0

(3) 9

(4) 0.11

50. If K_C for the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is 16, then K_C for the reaction, $NH_3 \rightleftharpoons \frac{1}{2}N_2 + \frac{3}{2}H_2$ will be

(1) 16

(2) 8

(3) 0.25

(4) 0.125

51. N_2 and H_2 are taken in molar ratio of 1 : 3. At equilibrium 50% of reaction has taken place, generating equilibrium pressure P. The pressure of NH_3 at equilibrium will be

(1) $\frac{P}{2}$

(2) $\frac{P}{3}$

(3) $\frac{P}{4}$

(4) $\frac{P}{5}$

52. A weak monoprotic acid HX has the dissociation constant 1×10^{-6} M. It forms a salt NaX on reaction with strong alkali. The degree of hydrolysis of 0.01 M solution of NaX is

(1) 0.01%

(2) 0.1%

(3) 0.001%

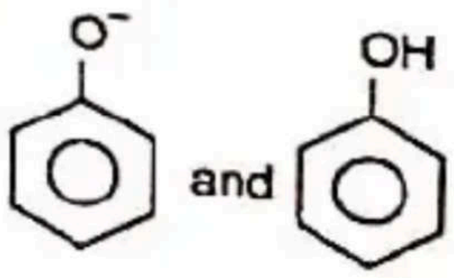
(4) 0.15%

53. Which among the following is a conjugate acid base pair?

(1) H_2CO_3 and CO_3^{2-}

(2) H_3O^+ and OH^-

(3) H_3PO_4 and HPO_4^{2-}

(4)  and

54. For an endothermic equilibrium $A(g) + B(g) \rightleftharpoons 2C(g) + D(g)$, the equilibrium will be shift in forward direction by

(1) Decreasing temperature

(2) Adding inert gas at constant pressure

(3) Adding catalyst

(4) Increasing pressure

55. If K_1 is the equilibrium constant at temperature T_1 and K_2 is the equilibrium constant at temperature T_2, and if $T_1 > T_2$ and reaction is exothermic then

(1) $K_2 = K_1$

(2) $K_2 > K_1$

(3) $K_1 > K_2$

(4) Can't be predicted

Einstein X369

the pH of the solution obtained by mixing 250 ml, 0.2 M CH_3COOH and 200 ml 0.1 M NaOH is (Given: pK_a of $CH_3COOH = 4.74$, log 3 = 0.48)

(1) 5.22

(2) 4.92

(3) 4.56

(4) 5.04

57. If solubility of AgCl(s) in water is 10^{-5} M then the solubility of AgCl in 0.1 M $AgNO_3$ will be

(1) 10^{-10} M

(2) 10^{-8} M

(3) 10^{-12} M

(4) 10^{-9} M

58. Which of the following would cause forward shift in the given reaction?

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \Delta H = -x$ kJ

(1) Increase of temperature

(2) Decrease of pressure

(3) Addition of inert gas at constant pressure

(4) Addition of HCl gas

59. For a weak base, pK_b is 4.8. If pOH of the solution of weak base and its salt with a strong acid is 5.8, the ratio of concentration of base to its salt with a strong acid will be

(1) 0.1

(2) 3

(3) 6

(4) 10

60. Select the correct expression

(1) $\Delta G = -\Delta G^\circ + RT \ln K$

(2) $\Delta G = \Delta G^\circ - RT \ln K$

(3) $\Delta G = \Delta G^\circ + RT \ln Q$

(4) $\Delta G^\circ = RT \ln K$

61. Equilibrium constant of a reaction depends on

(1) Temperature

(2) Pressure

(3) Volume

(4) Concentrations of reactants and products

62. When 50 mL of 0.1 M NH_4OH is mixed with 25 mL of 0.1 M HCl, then pH of the resulting mixture is [$pK_b = 4.74$]

(1) 4.74

(2) 9.26

(3) 7

(4) 10

63. pH of a solution of 0.1 M CH_3COONH_4(aq) is (given: $K_a(CH_3COOH) = K_b(NH_4OH) = 1.8 \times 10^{-5}$)

(1) 1

(2) 9

(3) 7

(4) 13

64. The pH of aqueous solution of salt of weak acid and weak base at 25°C is given by

(1) $pH = 7 + \frac{1}{2}(pK_a + pK_b)$

(2) $pH = 7 + \frac{1}{2}(pK_a - pK_b)$

(3) $pH = 7 - \frac{1}{2}(pK_a - pK_b)$

(4) $pH = 7 - \frac{1}{2}[pK_a + pK_b]$

65. The molar solubility of AlI_3 ($K_{sp} = 1 \times 10^{-12}$) in 0.1 M solution of KI will be

(1) 1×10^{-4} M

(2) 1×10^{-9} M

(3) 1×10^{-2} M

(4) 1×10^{-12} M

66. What amount of Ag_2CO_3 is dissolved in 250 mL of water? [$K_{sp} = 8 \times 10^{-12}$] (Gram atomic mass of Ag is 108 g mol^{-1})

(1) 8.69×10^{-3} g

(2) 1.25×10^{-4} g

(3) 2.5×10^{-4} g

(4) 6.2×10^{-4} g

67. The variation of equilibrium constant (K) with temperature is represented by

(1) $\left(\frac{d \ln K}{dT}\right)_p = -\frac{\Delta H}{RT}$

(2) $\ln K = \text{constant} + \frac{\Delta H}{RT}$

(3) $\ln K_2 - \ln K_1 = -\frac{\Delta H}{R}\int_{T_1}^{T_2} d\left(\frac{1}{T}\right)$

(4) $\log K = \text{constant} - \frac{\Delta H}{T}$

68. For the reversible reaction by

$PCl_5(g) + heat \rightleftharpoons PCl_3(g) + Cl_2(g)$

The equilibrium shifts in forward direction by

(1) Adding catalyst

(2) Increasing pressure

(3) Decreasing temperature

(4) Removing Cl_2

Einstein

69. Given : $\frac{1}{2}N_2 + \frac{1}{2}O_2 \rightleftharpoons NO$, $K_C = 5$
The value of equilibrium constant (K_C) for the reaction $N_2 + O_2 \rightleftharpoons 2NO$ will be

(1) 5

(2) 25

(3) 2.23

(4) 125

70. Which among the following is a conjugate acid base pair?

(1) H_2SO_4, SO_3

(2) H_3PO_4, PO_4^{3-}

(3) HNO_3, NO_3^-

(4) $HPO_2^-, H_2PO_3^-$

71. If $K_W = 10^{-12}$ at a temperature t°C then pH of pure water at t°C is

(1) 1

(2) 6

(3) 7

(4) 12

72. Hydrolysis of CH_3COONa increases when it is mixed with the solution of

(1) CH_3COOH

(2) NaOH

(3) HCl

(4) NH_4OH

73. pH of 0.05 M aqueous solution of weak acid HA ($K_a = 4 \times 10^{-4}$) is

(1) 4.70

(2) 2.35

(3) 3.39

(4) 1.30

74. If at T(K), ionic product of water is 10^{-13} then at what pH the solution will be acidic?

(1) 7

(2) 6.2

(3) 6.8

(4) 13

75. If pK_a of monobasic weak acid HA is 5 then the pH of 0.01 M NaA solution is

(1) 5

(2) 8.5

(3) 2.5

(4) 11.5

76. The reaction $A(g) \rightleftharpoons 2B(g) + C(g)$ has begun with 1 mol of A in 10 L closed vessel. When equilibrium is reached, the concentration of C is 0.08 M. The value of equilibrium constant(K_C) for this reaction is given by expression

(1) $\frac{0.08(0.16)^2}{0.2}$

(2) $\frac{0.08(0.16)^2}{0.92}$

(3) $\frac{0.16(0.08)^2}{0.02}$

(4) $\frac{0.16(0.08)^2}{0.92}$

77. For the equilibrium, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, if inert gas is added at constant pressure then

(1) Equilibrium will shift towards forward direction

(2) Equilibrium will remain unaltered

(3) Equilibrium will shift in backward direction

(4) Effect on equilibrium will depend on the type of inert gas used

78. If concentration of F^- in saturated solution of CaF_2 is x then solubility product of CaF_2 will be

(1) x^3

(2) $4x^3$

(3) $\frac{x^3}{2}$

(4) $\frac{x^3}{4}$

79. At t°C, K_W for water is 6.4×10^{-13}. The pH of water at t°C will be

(1) 6.91

(2) 5.90

(3) 6.09

(4) 5.15

80. When equal volumes of two solutions pH = 2 and pH = 3 are mixed together then the pH of resulting solution will be [log 5.5 = 0.74]

(1) 2.74

(2) 3.74

(3) 2.26

(4) 2.52

81. Which one of the following is a Lewis base?

(1) Ag^+

(2) $AlCl_3$

(3) PH_3

(4) BF_3

Einstein

(g) ⇌ 2B(g) + C(g) has begun with
in L closed vessel. When equilibrium
ion of C is 0.08 M. The value of
reaction is given by

of the pH of aqueous solution of 0.02 M dimethylamine is 12, then K_b of the dimethylamine is

(1) 1.5×10^{-3}

(2) 1×10^{-2}

(3) 1×10^{-4}

(4) 0.5×10^{-2}

83. The equimolar salt solutions having lowest pH is

(1) CH_3COONa

(2) CH_3COOK

(3) NH_4Cl

(4) NaCN

84. Conjugate base of H_3PO_4 is

(1) $H_2PO_4^-$

(2) HPO_4^{2-}

(3) PO_4^{3-}

(4) All of these

85. A weak acid having dissociation constant 10^{-6} is being titrated with aqueous NaOH solution. The pH of solution at one-third equivalence point of the acid will be

(1) 4.23

(2) 9.76

(3) 5.7

(4) 4

86. The unit of equilibrium constant Kp for the reaction, $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ is

(1) atm

(2) atm^2

(3) atm^{-1}

(4) atm^{-2}

87. Select the correct option regarding boric acid

(1) Arrhenius acid

(2) Tribasic acid

(3) Bronsted acid

(4) Lewis acid

88. The salt which undergoes cationic hydrolysis is

(1) CH_3COONa

(2) $(NH_4)_2SO_4$

(3) CH_3COOK

(4) NaCN

89. If for the reaction $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$, K_C is 2×10^{10}, then the Kp for the reaction at the same temperature T K is

(1) 2×10^{10}

(2) 2×10^{10} RT

(3) $\frac{2 \times 10^{10}}{RT}$

(4) $\frac{RT}{2 \times 10^{10}}$

90. pH of solution made by mixing 200 mL of 0.2 M CH_3COONa and 400 mL 0.1 M CH_3COOH is [$pK_a(CH_3COOH) = 4.7$]

(1) 3.2

(2) 4.7

(3) 5.2

(4) 6.8

Einstein

Botany

91. Identify the correct floral formula of a family to which belladonna belongs.

(1) $Br \oplus \male \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$

(2) $\oplus \male K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$

(3) $\% \female K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_1$

(4) $\oplus \male K_{2+2} C_{\cdot 4} A_{2+4} \underline{G}_{(2)}$

92. $\oplus \female K_{(5)} \overset{\frown}{C_{(5)}} A_0 \underline{G}_{(2)}$
is the floral formula of family
(1) Liliaceae
(2) Solanaceae
(3) Brassicaceae
(4) Fabaceae

93. When female reproductive part occupies the highest position and other parts are situated below it, the flower is called
(1) Hypogynous with inferior ovary
(2) Hypogynous with superior ovary
(3) Epigynous with inferior ovary
(4) Epigynous with superior ovary

94. Tendrils of pea differ from cucumber as the former
(1) Are sensitive structures
(2) Are modified leaves
(3) Coil around the nearby support
(4) Help the plant in climbing

95. In which of the given plants flowers are asymmetric?
(a) *Canna*
(b) *Cassia*
(c) Bean
(d) Mustard
(e) *Datura*
Choose the correct one(s).
(1) Only (b)
(2) Both (c) and (e)
(3) Both (d) and (e)
(4) Only (a)

96. 'Margins of petals overlap each other, but not in a ... pattern' is true for
(1) China rose
(2) Mustard
(3) Cassia
(4) Bean

97. Choose the **odd** one for axile placentation.
(1) Lemon
(2) Mustard
(3) Tomato
(4) China rose

98. Bicarpellary, obligately placed septum, axile placentation is found in flowers of
(1) Poaceae
(2) Liliaceae
(3) Solanaceae
(4) Fabaceae

99. The ovary is one chambered but become two chambered due to formation of false septum in _____ placentation.
(1) Marginal
(2) Axile
(3) Parietal
(4) Free central

100. Which of the given feature is **not** related to mustard?
(1) Superior ovary
(2) Variable length of filaments of stamens
(3) Parietal placentation
(4) Opposite phyllotaxy

101. The margin of the thalamus grows upward enclosing the ovary completely and getting fused with it, such condition of flower occurs in
(1) Guava, peach
(2) Cucumber, ray florets of sunflower
(3) Bittergourd, cucumber
(4) All except (1)

102. Which of the following is a modification of primary/tap root?
(1) Prop roots
(2) Stilt root
(3) Pneumatophores
(4) Root tuber of *Asparagus*

Einstein 1369

EinsteinX369

120. Match the following columns and select the correct option with correct combination.

a.	Mangrove	i)	Lemon
b.	Free central	ii)	Mustard
c.	Axile	iii)	Pea
d.	Parietal	iv)	Primrose

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(iv), c(i), d(ii)

121. Dialypetalous condition of androecium is seen in

(1) Citrus

(2) Pea

(3) China rose

(4) Solanum

122. Match the plant and its family having the following features.

(a) Androecium - adelphous

(b) Flower - actinomorphic and hypogynous

(c) Gynoecium - tricarpellary, bilocular, syncarpous

(d) Corolla - valvate aestivation

	Plant	Family
(1)	Sesbania	Brassicaceae
(2)	Cochicum	Liliaceae
(3)	Ashwagandha	Solanaceae
(4)	Trifolium	Fabaceae

(1) (1)

(2) (2)

(3) (3)

(4) (4)

123. In a floral formula, corolla is represented as $C_{(1-2+(2)}$. Such type of corolla is found in which of the following plants?

(1) Brassica

(2) Petunia

(3) Indigofera

(4) Asparagus

124. The term gamosepalous is used for

(1) Corolla when its components are free

(2) The components of corolla attached with gynoecium

(3) Sepals which are united with gynoecium

(4) Calyx when its components are united

125. Seed coat is membranous and fused with fruit wall in

(1) Gram

(2) Maize

(3) Bean

(4) Pea

126. The edible part of mango fruit is

(1) Endosperm

(2) Epicarp

(3) Mesocarp

(4) Epicarp

127. Match the columns and select the correct option

Column-I	Column-II
(a) Marginal Placentation	(i) Sunflower
(b) Variation in the length of filament	(ii) Gulmohur
(c) Zygomorphic flowers	(iii) Pea
(d) Swollen placenta	(iv) Mustard

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iv)	(iii)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iii)	(ii)	(i)	(iv)
(4)	(iv)	(iii)	(ii)	(i)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

128. Read the following statements of assertion and reason and choose the correct option.

Assertion (A): Inflorescence in the members of family Brassicaceae is said to be racemose.

Reason (R): In the members of family Brassicaceae, floral axis continues to grow and the flowers are borne laterally in an acropetal succession.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false statement

(4) Both (A) and (R) are false statements

129. A proteinaceous aleurone layer

(a) is present in maize grain.

(b) Consists of thick cells.

(c) Provides nourishment.

(d) Is a part of endosperm.

(1) Only (a) is correct

(2) Only (a) & (d) are correct

(3) Only (a), (c) & (d) are correct

(4) All (a), (b), (c) & (d) are correct

130. In maize seed, radicle is enclosed by a sheath known as

(1) Scutellum

(2) Coleorhiza

(3) Coleoptile

(4) Testa

Einstein x369

EinsteinX369

Zoology

136. Read the following statements related to the neural system in humans.
Statement-I : The electrical potential difference across the plasma membrane of a resting neuron is called the resting potential.
Statement-II : A nerve impulse is transmitted from one neuron to another through junctions called synapses.
Select the **correct** option.

(1) Statement I is correct but statement II is incorrect

(2) Statement II is correct but statement I is incorrect

(3) Both the statements I and II are correct

(4) Both the statements I and II are incorrect

137. The $Na^+ - K^+$ pumps transport how many Na^+ and K^+ ions per ATP spent for maintaining the polarised state when the neuron is **not** conducting any impulse?

(1) 2 Na^+ and 3 K^+

(2) Na^+ and 3 K^+

(3) 3 Na^+ and 2 K^+

(4) 2 Na^+ and K^+

138. In humans, the depolarisation of axonal membrane occurs due to rapid

(1) Efflux of Na^+ in the tissue fluid

(2) Efflux of K^+ in the tissue fluid

(3) Influx of Na^+ in the axoplasm

(4) Influx of K^+ in the axoplasm

139. Choose the **correct** statements w.r.t. midbrain
(a) It is located between hypothalamus of the forebrain and pons of hindbrain.
(b) Cerebral aqueduct passes through midbrain.
(c) It is a part of brain stem.
(d) The midbrain consists of taste association area

(1) (a) & (b) only

(2) (a), (b) & (c)

(3) (a), (b), (c) & (d)

(4) (a), (b) & (d)

140. Which of the following is required to cause depolarisation in a resting neuron?

(1) Rapid influx of K^+ into axoplasm

(2) Efflux of Na^+ from axoplasm

(3) Influx of K^+ and efflux of Na^+ simultaneously

(4) Rapid influx of Na^+ into axoplasm

141. When a neuron is not conducting any impulse, the axonal membrane is

(1) Completely impermeable to Na^+ ions

(2) Permeable to negatively charged proteins present in the axoplasm

(3) Positive on inner surface while negative on outer surface

(4) More permeable to K^+ ions

142. Presence of which of the following is **not** an exclusive feature of a neuron?

(1) Plasma membrane

(2) Nissl's granules

(3) Dendrites

(4) Axon

143. What is true about action potential across the plasma membrane of a neuron?

(1) It is electrical potential difference across the plasma membrane during resting state

(2) The membrane at the point of applied stimulus becomes depolarised

(3) It is the result of rapid efflux of Na^+

(4) Inner side of plasma membrane becomes negatively charged due to this

144. Differential distribution of which ions across axolemma chiefly help to maintain resting membrane potential?

(1) Mg^{2+} and Ca^{2+}

(2) Na^+ and K^+

(3) Cl^- and Na^+

(4) K^+ and Mg^{2+}

145. Select the **incorrect** statement about electrical synapses

(1) They are rare in our system

(2) Impulse transmission is faster as compared to a chemical synapse

(3) Pre and post synaptic membranes possess a wide gap between them

(4) Neurotransmitters are absent at these synapses

Einstein X369

A neuron's resting membrane potential is established and maintained by all of the following **except**

(1) Differences in both ion concentrations & electrical gradients

(2) The plasma membrane's higher permeability to Na^+ because of the presence of numerous Na^+ channels

(3) There are numerous large, non-diffusible anions in the cytosol

(4) Sodium-potassium pumps that help to maintain the proper distribution of sodium & potassium

147. During repolarisation of a neuron, select the ion which moves outside from intracellular fluid of axon by voltage gated channels.

(1) K^+

(2) Na^+

(3) Ca^{+2}

(4) Cl^-

148. Action potentials temporarily disrupt the polarisation of a neuron, resulting in depolarisation. The action potential is determined largely by sudden

(1) Efflux of Na^+ in the tissue fluid

(2) Efflux of K^+ in the tissue fluid

(3) Influx of Na^+ in the axoplasm

(4) Influx of K^+ in the axoplasm

149. An electrical synapse differs from a chemical synapse as

(1) Former is unidirectional while latter is bidirectional

(2) Latter has wider synaptic clefts than former

(3) Former employs use of neurotransmitters

(4) Latter utilises presence of gap junctions to release chemicals into clefts

150. Read the following given statements and choose the **correct** option.
Statement I : Corpora quadrigemina is the tract of nerve fibres which connect left and right cerebral hemispheres.
Statement II : Midbrain, pons and medulla oblongata form the brain stem.

(1) Statement I is correct.

(2) Statement II is correct.

(3) Both statements I and II are correct.

(4) Both statements I and II are incorrect.

151. The ions which stimulate the fusion of synaptic vesicles with pre-synaptic membrane are

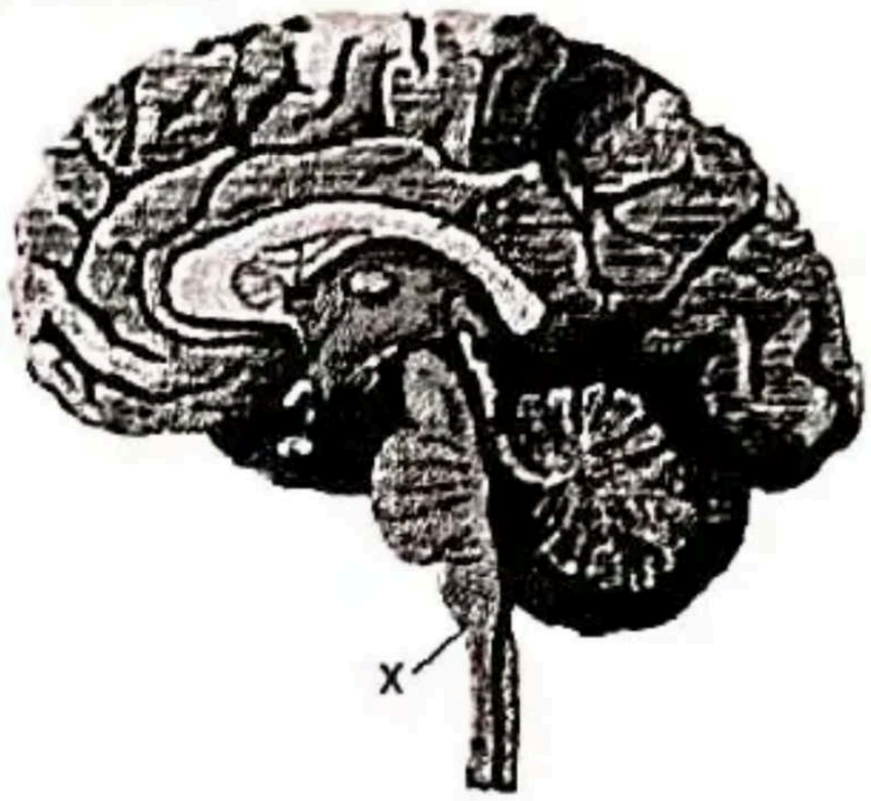
(1) PO_4^{3-}

(2) H^+

(3) Ca^{++}

(4) Mg^{++}

152. Given below is the diagram of human brain. Select the **Incorrect** statement regarding function of structure labelled as 'X'.



(1) Helps in understanding speech by interpreting sound

(2) Regulates respiratory rhythm

(3) Controls cardiovascular reflexes

(4) Controls gastric secretions

153. A : The imbalance in concentration of Na^+, K^+ and proteins generates resting potential.
R : To maintain the unequal distribution of Na^+ and K^+, $Na^+ - K^+$ pump moves ions activity along the concentration gradient.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

154. Application of threshold stimulus at a point on the resting membrane will lead to

(1) Switching on of $Na^+ - K^+$ pump

(2) Opening of voltage gated Na^+ channels

(3) Rapid diffusion of K^+ from ECF to axoplasm

(4) Rapid diffusion of Na^+ from axoplasm to ECF

155. Choose the **Incorrect** statement w.r.t. electrical synapse.

(1) The membranes of pre-and post-synaptic neurons are in very close proximity.

(2) Transmission of impulses across this synapse is very similar to impulse conduction along a single axon.

(3) Electrical current can flow directly from one neuron into other across these synapses.

(4) Electrical synapses are very common in our body.

EinsteinX369

[illegible text and blank space]

156 Which impulses in skeletal fibre result in

(1) Resting potential

(2) Depolarisation

(3) Action potential

(4) Repolarisation

157 Which ions are responsible for nerve impulse conduction?

(1) Ca^{2+} ions

(2) Na^{+} ions

(3) K^{+} ions

(4) Mg^{2+} ions

158 Arrange them in their correct sequence from outer to inner side

(1) a, b, c, d

(2) b, a, c, d

(3) a, b, d, c

(4) b, c, a, d

159 Which of the following is not a part of chemical synapse?

(1) Synaptic knobs

(2) Pre-synaptic membrane

(3) Synaptic cleft

(4) Post synaptic membrane

160 Depolarisation of nerve fibre is characterised by the

(1) Rapid influx of K^{+}

(2) Rapid influx of Na^{+}

(3) Rapid influx of Na^{+} & K^{+}

(4) Rapid influx of Ca^{2+}

161 All of the following are parts of brain stem except

(1) Hypothalamus

(2) Midbrain

(3) Pons

(4) Medulla oblongata

162 A neural synapse is formed by the

(1) Pre-synaptic membrane and synaptic cleft without post synaptic neuron

(2) Synaptic cleft and post synaptic membrane only

(3) Synaptic vesicle and dendrites

(4) Membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by synaptic cleft

163 Select the incorrect statement w.r.t. human neural system

(1) It is completely enclosed in skull

(2) Somatic neural system relays impulses from the CNS to skeletal muscles

(3) Visceral neural system comprises of the whole complex of nerves, fibres, ganglia and plexus

(4) The PNS comprises of all the nerves of the body associated with brain, brain and spinal cord

164 Following are the statements about potential
(a) Resting potential is due to the differences in permeability of K^{+} and P^{-}
(b) Depolarisation occurs when the membrane becomes more permeable to Na^{+}
(c) Repolarisation is due to the outward movement of K^{+}
(d) Hyperpolarisation is caused due to the inward movement of Na^{+}
Choose the correct answer from the options given below

(1) a and b are correct

(2) b and c are correct

(3) c and d are correct

(4) b and d are correct

165 Depolarisation of neuromuscular junction is caused due to

(1) Closure of voltage gated Na^{+} and K^{+} channels

(2) Opening of voltage gated Na^{+} channels

(3) Opening of voltage gated K^{+} channels

(4) Closure of K^{+} leaky channels

166 In humans, unipolar neuron is usually found in the

(1) Cerebral cortex of adult

(2) Embryonic stage

(3) Retina of human eye

(4) Organ of Corti in inner ear

167 Select the correct match

(1)	Hypothalamus	–	Coordinating centre for sensory and motor signalling and also known as relay centre
(2)	Corpus callosum	–	Tract of nerve fibres that connects two cerebellar hemispheres
(3)	Pons	–	Regulates the duration of inspiration
(4)	Medulla oblongata	–	Regulates urge for eating and drinking

(1) (1)

(2) (2)

(3) (3)

(4) (4)

168 Na^{+}–K^{+} ATPase maintains electro-chemical gradient across the resting membrane. If this pump is blocked, what is going to happen?

(1) Na^{+} inside the nerve will increase

(2) K^{+} inside the nerve will increase

(3) Na^{+} outside the nerve will increase

(4) Both Na^{+} and K^{+} outside the nerve will increase

Einstein369

Assertion (A) : Association areas of the cortex of forebrain are responsible for intersensory associations, memory and communications.
Reason (R) : Association areas are regions of cortex which are neither clearly sensory nor motor in function.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of the (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

169. Read the functions listed below
a. Production of stimulus
b. Detection of stimulus
c. Reception of stimulus
d. Transmission of stimulus
Select the option that represents the correct functions of a neuron.

(1) a and c only

(2) b, c and d only

(3) a, b and d only

(4) a, b, c and d

170. Arrange the following cranial meninges from inside to outside in human brain and select the **correct** option.
(a) Arachnoid
(b) Pia mater
(c) Dura mater

(1) (b) → (a) → (c)

(2) (a) → (b) → (c)

(3) (c) – (a) → (b)

(4) (c) – (b) – (a)

171. Which part of the hindbrain regulates cardiovascular reflexes?

(1) Medulla oblongata

(2) Pons

(3) Cerebrum

(4) Hypothalamus

172. A special neural centre in the brain which can moderate cardiac functions through ANS is present in

(1) Cerebellum

(2) Cerebrum

(3) Medulla oblongata

(4) Pons varolii

173. Read the following statements and choose the **correct** option.
Statement A : Dendrites conduct nerve impulses towards the cell body or cyton.
Statement B : From cyton, the axon conducts the nerve impulse towards the nerve endings.

(1) Statements A is correct but B is incorrect

(2) Statements B is correct but A is incorrect

(3) Both statements are correct

(4) Both statements are incorrect

174. 'Association areas' are responsible for complex functions like intersensory associations, memory and communication. These association areas are majorly the part of

(1) Corpora quadrigemina

(2) Cerebellum

(3) Cerebral cortex

(4) Medulla oblongata

175. In humans, the part of a neuron responsible for receiving stimuli in a chemical synapse is/are

(1) Dendrites

(2) Nissl's granules

(3) Axon

(4) Axon terminals

176. A unipolar neuron

(1) Does not have soma or body

(2) Possesses a single dendrite and single axon process

(3) Does not have dendrites

(4) Has numerous axons

177. All of the following is under regulation of sympathetic and parasympathetic nervous system, except

(1) Heart rate

(2) Digestive tract

(3) Learning

(4) Excretion

178. Nerve fibres of tracts present in white matter of the brain are covered with

(1) Corpus callosum

(2) Neurilemma

(3) Myelin sheath

(4) Gray matter

Einstein Academy

179. Nerve impulse is also termed as

(1) Chemical synapse

(2) Resting potential

(3) Electrical synapse

(4) Action potential

180. Function of bipolar retinal cells is to

(1) Carry signals from one photoreceptor cell to another

(2) Transmit the output from photoreceptor cells

(3) Absorb all wavelengths of light not captured by photoreceptors

(4) Bend light and focus it on macula lutea

EinsteinX369